

Hooman Ramezani

hooman125@gmail.com | www.hoomanramezani.com | github.com/HoomanRamezani

EDUCATION

University of Toronto

Toronto, ON

MASc in Machine Learning & Operations Research

2022 – 2024

- Thesis: Transformers for clinical workloads; LLM and ViT models with multimodal data for lung cancer treatment

University of Waterloo

Waterloo, ON

BASc in Systems Design Engineering

2017 – 2022

- GPA: 3.7 / 4.0 — Coursework: Intro. Deep Learning, Intro. Machine Learning, Pattern Recognition, Neuroscience

EXPERIENCE

ML Specialist Solutions Architect

Jan 2025 – Present

Nebius Group (Neocloud)

Remote

- Client-facing SA specializing in GPU performance, unblocking large-scale training workloads and assisting with migration to Blackwell architecture
- Profile and optimize client workloads on B300 / GB300, raise MFU, and stand up GPU clusters (1000+ GPUs)
- Token Factory, tuning inference engines like vLLM and TRT-LLM and deploying optimized production endpoints for clients
- Co-authored joint blog with Meta PyTorch team on accelerating MoE training on 256 B200 GPUs with MXFP8 and DeepEP, presented at NVIDIA GTC 2026

Machine Learning Researcher

May 2023 – Dec 2024

University Health Network

Toronto, ON

- First-author ‘LN-Transformer’ paper for lung nodule segmentation, achieving DiceC of 94.2% (SOTA)
- Built **multi-modal LLM** with unstructured text, CT slices for treatment recommendations with **84%** concordance

Machine Learning Engineer

Sept. 2022 – April 2023

Advanced Micro Devices (AMD)

Toronto, ON

- Optimized **LLM inference** on AMD CPUs via QAT, post-quantization, pruning, and knowledge distillation, optimized GPU kernel execution to reduce inference latency by **80%** and maximize throughput
- Benchmarked against **Triton** Inference Server with **TensorRT** backend on k8s, leveraging dynamic batching

Machine Learning Engineer

Jan. 2022 – April 2022

DarwinAI (Acquired by Apple)

Remote

- Delivered medical vision model to Pfizer for advanced liver fibrosis diagnosis, achieving **91% sensitivity**
- Built end-to-end ML solutions to clients with MLOps CI/CD pipelines, enabling continuous retraining, monitoring

RESEARCH // PROJECTS

LN-Transformer: Lung Nodule Transformer for Sparse CT Segmentation

2025

- Published a novel two-stage transformer for lung nodules segmentation featured in CVPR
- Strongest model on benchmark dataset **Dice 91.4%**, **F1 94.2%**, includes Meta SAM, DETR architectures

Enhanced Robotic Grasping with PointNet | LiDAR, Robotics, 3D Computer Vision

2022

- Trained PointCloud grasp proposition neural-net achieving 87% grasp success rate, identify optimal grasp

TECHNICAL SKILLS

Programming: Python, C/C++, Java, SQL, JavaScript

ML & DL: PyTorch, TensorFlow, Hugging Face, LoRA / PEFT, RAG, AI Agents, TensorRT, ONNX, Triton, CUDA

Distributed Systems: PyTorch DDP, FSDP, Spark, Kafka, Airflow, Docker, Kubernetes, Slurm

Cloud & MLOps: AWS (S3, EC2, Lambda, SageMaker), GCP, MLflow, Prometheus, Redis, Flask, Git